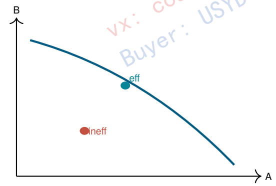
W1 PPF, TRADE, BASICS

Economics: choice under scarcity; **opp cost** = value of next best alt, not accounting cost.

opp cost of $x = \Delta y/\Delta x$ along PPF

If PPF bowed out: increasing OC from specialized resources; straight: constant OC.
PPF reading: feasible on/inside; efficient on frontier; growth shifts frontier; trade changes consumption possibilities not production PPF.
Advantage: absolute = higher productivity; comparative = lower OC. Gains from trade if each specializes where OC lower and terms of trade lie between OCs.

Signal	Interpretation	Pitfall
Inside PPF	inefficient/unemployed	not impossible
On PPF	productive efficient	may be alloc inefficient
Beyond PPF	infeasible now	possible after growth



Exam moves: compare slopes for CA; convert output/hour into OC; if both gain, TOT must be between OCs.
Marginal thinking: choose next unit iff MB $\ge$ MC; sunk costs ignored; incentives change behavior at margin. Example: if A gives up 2B for 1A and partner gives up 5B for 1A, A has CA in A; trade price between 2 and 5 B per A.
Specialization algorithm: build OC table; mark lower OC; check if proposed trade price is above seller OC and below buyer OC; translate gains into extra units.
Normative vs positive: positive claims testable/descriptive; normative claims value-based/prescriptive. Efficiency is positive once criterion fixed; fairness usually normative.

When a PPF rotates outward around one intercept, productivity improved in one good only; when both intercepts expand, broad growth or resource increase.

Edge case: one person can have absolute advantage in both goods but cannot have comparative advantage in both goods unless OCs tie; ties imply no strict CA.

W2 DEMAND AND WTP

Demand: quantity buyers are willing/able to buy at each price; inverse demand is marginal willingness to pay (MWTP) for each unit.

$Q_d = a - bP \Leftrightarrow P = a/b - (1/b)Q$

Movement vs shift: price change moves along demand; income/tastes/substitutes/complements/expectations/number buyers shift curve.
Goods: normal $I\uparrow \Rightarrow D\uparrow$; inferior $I\uparrow \Rightarrow D\downarrow$; substitutes $P_s\uparrow \Rightarrow D_x\uparrow$; complements $P_c\uparrow \Rightarrow D_x\downarrow$.

Individual MV schedule aggregates horizontally for private goods: at price P, sum quantities demanded.

Change	Demand shift	Eq effect if S fixed
Income up normal	right	$P, Q\uparrow$
Sub price up	right	$P, Q\uparrow$
Complement price up	left	$P, Q\downarrow$
Expected future price up	right now	$P, Q\uparrow$

Elasticity warning: slope $\neq$ elasticity. Same line has changing elasticity.

$E_d = \frac{\%\Delta Q}{\%\Delta P} = \frac{dQ}{dP}\frac{P}{Q}$; $|E_d| > 1$ elastic, $|E_d| < 1$ inelastic

Revenue test: price rise increases TR iff demand inelastic; lowers TR iff elastic; max TR at unit elastic for linear demand.
Consumer surplus: area below demand/MWTP above price; for linear demand triangle $\frac{1}{2}(P_{max} - P^*)Q^*$. MCQ trap: a higher WTP buyer may still receive same good price; surplus is WTP minus price, not WTP itself.
Discrete MV table: buy all units with $MV > P$; CS $= \sum(MV_i - P)$ for purchased units. If supply cost c per unit, efficient units satisfy $MV_i \ge c$.

Demand choke price: price at which $Q_d = 0$; useful for CS triangles and monopoly inverse demand.

Elasticity determinants: more substitutes, longer time horizon, higher budget share, luxury status $\Rightarrow$ more elastic; necessity/addiction $\Rightarrow$ less elastic.
Trap: if $Q_d = 6 + 3P$, it violates law of demand unless context is supply or typo; demand should slope down in standard models.

W3 PRODUCTION AND COSTS

Production fn: $Q = f(K, L)$ maps inputs to max output. Short run: some fixed input; long run: all variable.

$MP_L = \Delta Q/\Delta L$, $AP_L = Q/L$

Diminishing MP: with fixed capital, extra labor eventually adds less output; drives rising MC.
Costs: fixed FC independent of q; variable $VC(q)$; total $TC = FC + VC$; marginal $MC = \Delta TC/\Delta q$.

$ATC = TC/q = AFC + AVC$, $AFC = FC/q$, $AVC = VC/q$, $MC = \frac{dTC}{dq}$

MC crosses AVC and ATC at their minima; if $MC < ATC$, ATC falls; if $MC > ATC$, ATC rises.

Output	TC	MC	ATC
q	$FC + VC$	slope TC	TC/q
q inc	AFC falls	often U-shape	U-shape
Decision	profit	produce unit	avg cost

Scale: economies if LRATC falls as $q\uparrow$; diseconomies if rises; constant if flat. Returns to scale: $f(tK, tL)$ vs $tf(K, L)$.
Past-paper style: given hourly output and OC of time, include both explicit input costs and implicit OC in TC. If price P, competitive firm chooses unit where $P \ge MC$ and profit $TR - TC$.
Isoquant: input bundles for same output; slope $MRTS = MP_L/MP_K$ in absolute value; convex when diminishing MRTS. Isocost: $C = wL + rK$, slope $-w/r$; cost min where $MRTS = w/r$.
Table workflow: compute VC, TC, then $MC = \Delta TC/\Delta Q$ between rows; $ATC = TC/Q$ at row; never average marginal costs to get ATC.

If $TC = F + cq + dq^2$, then $MC = c + 2dq$, $AVC = c + dq$, $ATC = F/q + c + dq$. Shutdown uses AVC, exit uses ATC.

Accounting vs economic profit: economic profit subtracts explicit and implicit costs; zero economic profit means normal return included, not zero accounting earnings.

Cost-curve diagnosis: rising AP of labor corresponds to falling AVC when wage fixed; falling MP corresponds to rising MC.

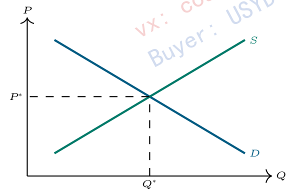
W4 SUPPLY AND DEMAND

Supply: quantity sellers willing/able at each price; inverse supply often MC for competitive producers. Shift vs movement mirrors demand logic.

Equilibrium solves $Q_d(P) = Q_s(P)$ or inverse $P_D(Q) = P_S(Q)$. At $P > P^*$ surplus; at $P < P^*$ shortage.

Shock algebra: (1) identify shifted curve, (2) solve new intersection, (3) compare P, Q, (4) explain ambiguity if both curves move opposite price directions.

Shock pair	Q^*	P^*
$D\uparrow, S\uparrow$	rises	ambiguous
$D\downarrow, S\downarrow$	falls	ambiguous
$D\uparrow, S\downarrow$	ambiguous	rises
$D\downarrow, S\uparrow$	ambiguous	falls



Price controls: binding ceiling below P^* causes shortage; binding floor above P^* causes surplus. Non-binding control has no effect.
Comparative statics: never say demand increases when quantity demanded rises due to price fall; that is movement along demand.
Linear solve example: if $Q_d = 100 - 2P$, $Q_s = 20 + 2P$, then $P^* = 20$, $Q^* = 60$; price floor 25 gives $Q_d = 50$, $Q_s = 70$, surplus 20.

A shortage/surplus is measured at the controlled price, not at the old equilibrium quantity.

Black market risk: binding ceilings create rationing queues and quality cuts; binding floors create storage/destruction/government purchase costs.
Ambiguous shock: when both demand and supply shift, one variable may be definite and the other ambiguous; use graph plus algebra signs.

W5 SURPLUS AND ELASTICITY

Total surplus: $TS = CS + PS$; efficient competitive outcome maximizes TS when no externalities/market power.

$CS = \int_0^{Q^*} P_D(q)dq - P^*Q^*$, $PS = P^*Q^* - \int_0^{Q^*} P_S(q)dq$

Deadweight loss is lost gains from mutually beneficial trades not made, or inefficient trades made.
Price elasticity demand: $E_d = (dQ/dP)(P/Q)$; supply: $E_s = (dQ_s/dP)(P/Q)$. Use absolute value for demand elasticity language.
Tax/subsidy incidence preview: less elastic side bears more burden/gains more from subsidy. Perfectly inelastic side bears all tax; perfectly elastic side bears none.

Elasticity	Formula cue	Effect
PED	$dQ/dP \cdot P/Q$	TR, incidence
Income	$\%\Delta Q/\%\Delta I$	normal/inferior
Cross	$\%\Delta Q_x/\%\Delta P_y$	sub/comp
Supply	$dQ_s/dP \cdot P/Q$	burden split

Midpoint: $E = \frac{\Delta Q/\bar{Q}}{\Delta P/\bar{P}}$ avoids base dependence.
Efficiency vs equity: surplus maximization says nothing about fairness; policy answers often need both. Example: if $P_D = 50 - Q$, $P_S = 10 + Q$, eq $Q = 20$, $P = 30$, CS=PS=200.

With linear demand/supply, surplus triangles use vertical height times horizontal base; do not multiply by old price unless rectangle revenue/cost is requested.

DWL from output distortion: find efficient Q_E and policy Q_P; area between demand and supply over missing/excess units.
Perfectly inelastic demand: vertical demand; quantity unchanged after tax, no DWL if no output change, buyers bear burden. Perfectly elastic demand: horizontal, sellers bear tax if supply upward.
Elasticity and tax revenue: raising a tax can reduce revenue when quantity response is large; maximum revenue not same as maximum welfare.

W6 EQ AND PERFECT COMP

Perfect competition: many buyers/sellers, homogeneous product, free entry/exit, price takers, full info. Firm faces horizontal demand at market price.

Profit max rule: produce where $P = MR = MC$ on upward MC, if price covers shutdown threshold.

Short run: produce if $P \ge$ min AVC; shut down if $P <$ min AVC; profit may be negative if $AVC \le P < ATC$.
Long run: enter if economic profit > 0; exit if profit < 0; LR eq $P =$ min ATC if identical firms/free entry.

$\pi(q) = Pq - TC(q)$, $MR = P$, $\pi > 0 \Leftrightarrow P > ATC$ at chosen q

Condition	SR action	LR action
$P < AVC$	shut down	exit if persists
$AVC \le P < ATC$	produce loss	exit
$P = ATC$	normal profit	stay
$P > ATC$	profit	entry erodes

Industry supply: SR sums firm MC curves above AVC; LR can be flat (constant-cost industry) or upward if input prices rise.
Trap: fixed cost affects profit, not SR output decision after entry; sunk cost should not determine marginal production.
Firm calculation: solve $P = MC(q)$; check shutdown $P \ge AVC(q)$; compute $\pi = Pq - TC(q)$; market quantity $Q = Nq$ if N identical firms.

Competitive supply curve is MC above AVC in SR, not the whole MC curve. At prices below AVC the supplied quantity is zero.

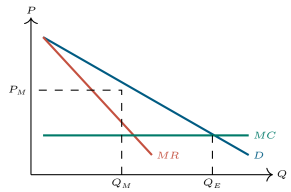
Entry effect: positive profit attracts entry, shifts market supply right, lowers price; losses cause exit, shifts supply left, raises price.
Welfare: competitive equilibrium is efficient only if price equals marginal social cost and marginal social benefit; externality breaks this.

W7 MONOPOLY I

Monopoly: single seller faces market demand; chooses Q then charges max WTP from demand. Market power from barriers: legal, natural monopoly, control input, network effects.

For inverse demand $P = a - bQ$: $TR = aQ - bQ^2$, $MR = a - 2bQ$. Profit max: $MR = MC$, then price from demand.

Monopoly diagram: demand above MR; choose Q_M where MR=MC; price $P_M = P_D(Q_M)$; efficient Q_E where $P_D = MC$.



$\pi = (P_M - ATC(Q_M))Q_M$; $DWL = \frac{1}{2}(P_M - MC)(Q_E - Q_M)$ if linear/constant MC

Markup: $(P - MC)/P = -1/E_d$ at optimum; monopoly operates on elastic part of demand. If demand inelastic, lower Q/raise P increases revenue and cuts cost.
Exam algebra: if $Q = 150 - 2P$, invert $P = 75 - 0.5Q$, so $MR = 75 - Q$, not $150 - 4P$.
Monopoly solve sequence: invert demand; compute $TR = P(Q)Q$; derive MR; set $MR = MC$; check demand price; calculate CS, PS/profit, DWL.

Revenue-max output solves $MR = 0$; profit-max output solves $MR = MC$. They coincide only when $MC = 0$.

Market power source: monopoly can be legal (patent), technical (natural monopoly), strategic (entry deterrence), or demand-side (network effects).
No supply curve: monopoly has no supply curve independent of demand because chosen quantity depends on MR, which depends on demand shape.

W8 MONOPOLY II

Price discrimination: charge different prices not justified by cost; needs market power, ability to segment/identify WTP, and prevent resale.
Degrees: first-degree perfect PD captures all CS; second-degree menu/quantity pricing; third-degree group pricing.

Third-degree rule: allocate output so $MR_1 = MR_2 = MC$; lower price in more elastic market.

Two-part tariff: fixed fee F plus per-unit price p. If identical consumers, set $p = MC$, fee extracts CS; with heterogeneity, trade off participation and usage distortion.

Policy	Quantity	Surplus distribution
Uniform monopoly	low	DWL, profit
Perfect PD	efficient	firm captures CS
Two-part identical	efficient	fee captures CS
Bundling	depends	screens WTP

Natural monopoly: high fixed cost, low MC; ATC declining over relevant range. Marginal cost pricing may require subsidy; average cost pricing covers cost with smaller output than efficient.
Regulation compare: price cap can reduce monopoly price but may lower quality/investment if too tight; lump-sum tax affects profit not output, per-unit tax raises MC and lowers Q.
Lerner intuition: high elasticity limits markup; no demand substitutes gives more pricing power.
Perfect PD welfare: sells every unit with $WTP \ge MC$ so no DWL, but CS transferred to producer; efficiency and equity diverge sharply.

Group pricing: if $|E_A| < |E_B|$, group A faces higher markup/price, all else equal.

Bundling: profitable when valuations are negatively correlated; can smooth WTP differences and increase extraction.
Two-part tariff pitfall: setting usage price above MC creates DWL; fixed fee extracts surplus without distorting usage if participation remains.

W9 GAME THEORY, OLIGOPOLY

Game: players, strategies, payoffs, timing, info. **Dominant strategy**: best regardless of opponent. **Nash equilibrium**: each strategy is best response to others.
Find NE: circle best responses by row player for each column and column player for each row; mutual circles are NE.

Game	Feature	Prediction
Prisoner's dilemma	dominant defect	inefficient NE
Coordination	multiple NE	expectations matter
Battle sexes	coord + conflict	multiple pure NE
Chicken	avoid mutual aggression	mixed possible

Sequential games: use game tree and backward induction; subgame perfect equilibrium rules out non-credible threats.

Oligopoly interdependence: each firm's best output/price depends on rivals. Cartel maximizes joint profit but faces cheating incentives.

Cournot: choose quantities; more firms $\Rightarrow$ price falls toward competitive. **Bertrand**: choose prices; identical products and no capacity constraints can yield $P = MC$.
Exam trap: no dominant strategy does not imply no NE. More than one NE often appears in coordination games like meeting locations.

	Left	Right
Up	A	B
Down	C	D

Mixed intuition: when no pure dominant strategy, equilibrium may randomize so opponent is indifferent; do not average payoffs mechanically.

Collusion problem: joint monopoly price raises industry profit, but each firm may gain by undercutting/expanding if punishment is weak.

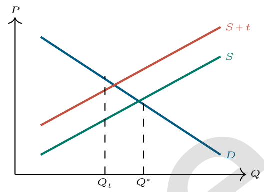
Repeated game: cooperation easier with high patience, clear monitoring, credible punishment, and stable firms; harder with many firms or volatile demand.
Sequential trap: a threat is credible only if optimal when the decision node is reached; backward induction checks this.

W10 TAX, SUBSIDY, REG

Per-unit tax: wedge between price buyers pay and sellers receive: $P_b = P_s + t$. Legal incidence irrelevant to economic incidence.

Solve tax: $Q_d(P_b) = Q_s(P_s)$ and $P_b - P_s = t$. Revenue $= tQ_t$; DWL $= \frac{1}{2}t(Q^* - Q_t)$ for linear curves.

Incidence: buyers' burden $P_b - P^*$; sellers' burden $P^* - P_s$; less elastic side bears more. Subsidy wedge $P_s - P_b = s$; quantity rises, fiscal cost sQ_s, DWL possible from overproduction.



Price ceiling: if binding, $Q_s < Q_d$, shortage, nonprice rationing, possible black markets. **Price floor**: if binding, $Q_s > Q_d$, surplus, wasted resources.
Tax sign trap: a tax on buyers shifts demand down by t; a tax on sellers shifts supply up by t; final wedge same.
Quota: quantity restriction can create quota rents; if licenses scarce, rent equals price gap times quota.
Subsidy algebra: $P_s - P_b = s$; solve $Q_d(P_b) = Q_s(P_s)$. Buyer benefit $P^* - P_b$, seller benefit $P_s - P^*$, fiscal cost sQ.

Per-unit tax with linear curves: if pre-tax Q^* and post-tax Q_t, DWL triangle base $Q^* - Q_t$, height t.

Ad valorem tax: percentage of price; wedge proportional to transaction price, not fixed dollars per unit.
Regulation answer template: state binding/non-binding, quantity traded, who gains/loses, DWL, and unintended effects (quality, queues, evasion).

W11 EXTERNALITIES

Externality: uncompensated effect on third party. Negative production: $MSC = MPC + MEC$; positive consumption: $MSB = MPB + MEB$.

Efficient quantity solves $MSB = MSC$, not private $MPB = MPC$ when externality exists.

Pigouvian tax: for negative externality, tax per unit equal to MEC at efficient quantity; aligns private MC with social MC. Subsidy for positive externality equals marginal external benefit at efficient quantity.

Case	Market outcome	Correction
Neg production	overproduce	tax/trade permits
Pos production	underproduce	subsidy/R&D support
Neg consumption	overconsume	tax/info rules
Pos consumption	underconsume	subsidy/provision

Coase theorem: if property rights clear and transaction costs low, bargaining can reach efficient allocation regardless of initial rights; distribution depends on rights.
Limit: many parties, asymmetric info, high transaction costs, wealth effects, enforcement problems.

$DWL_{\text{neg ext}} = \int_{Q_E}^{Q_M}(MSC - MSB)dQ$

Tradable permits: cap fixes quantity; permit price emerges; cost-effective if firms with low abatement cost abate more.
Negative externality graph: private market $MPB = MPC$ gives Q_M; social optimum $MSB = MSC$ gives lower Q_E; tax shifts MPC to MSC if exactly calibrated.

Corrective tax equals marginal external cost at Q_E, not necessarily average damage and not necessarily damage at Q_M.

Policy comparison: command-control can hit target but may ignore cost differences; permits/taxes use decentralized cost information; liability works when victims and harms identifiable.
Positive externality: education/vaccination examples; market underconsumes because private MB below social MB. Correct with subsidy/information/provision.

W12 PUBLIC GOODS, COMMONS

Rivalry: one person's use reduces availability. **Excludability**: possible to prevent non-payers.

	Excludable	Non-excludable
Rival	private good	common resource
Non-rival	club good	public good

Private goods: horizontal sum of demand. Public goods: vertical sum of MWTP, since same quantity consumed by all.

$MSB_{\text{public}}(Q) = \sum_i MB_i(Q)$; $Q^* : MSB(Q) = MC(Q)$

Free rider problem: non-excludability weakens voluntary payment; market underprovides public goods.
Common resources: rival + non-excludable; overuse/tragedy commons because private user ignores congestion/depletion cost on others. Policies: quotas, permits, property rights, access fees, community norms.
Public good calculation: convert each individual's inverse demand/MV at common Q; add vertically; solve $\sum MB_i = MC$. If n identical consumers $MB = a - bQ$, then $MSB = n(a - bQ)$.
Past-paper style: parks/fireworks/public radio. If individual $MV = 200 - Q$ and 500 people, $MSB = 100000 - 500Q$; efficient where $MSB = MC$.
Club goods: congestion after capacity; efficient membership/fee can finance provision when exclusion is feasible.
Lindahl intuition: personalized prices can finance public good if each pays according to marginal benefit, but information/free-riding makes implementation hard.

For common resources, private marginal cost excludes congestion/depletion cost; efficient use requires user fee or quota equal to external cost.

Exam examples: clean air public good-ish/non-excludable; fishery common resource; toll road club good if uncongested, common/club hybrid if congested; sandwich private good.
Answer discipline: classify by rivalry/excludability first, then predict market failure, then choose instrument and one limitation.
Integrated calculation bank:

Task	First equation	Then report
Eq market	$Q_d = Q_s$	P^*, Q^*
Tax	$P_b - P_s = t$	burden, rev, DWL
Monopoly	$MR = MC$	Q_M, P_M, π, DWL
Public good	$\sum MB_i = MC$	Q^*, free ride
Externality	$MSB = MSC$	tax/subsidy

Graph labelling checklist: axes P, Q; curves D/S or MSB/MSC; equilibrium coordinates; policy wedge/control; shaded CS/PS/DWL only if asked. Missing labels often lose marks even with right algebra.

Consumer vs producer surplus after tax: use price actually paid/received, not the statutory side. CS uses P_b; PS uses P_s.

Inverse demand shortcut: if $Q = a - bP$, then $P = a/b - (1/b)Q$, $MR = a/b - (2/b)Q$. If $P = a - bQ$, then $MR = a - 2bQ$.
Unit consistency: if Q is thousands, revenue/profit units follow; if price is per 1,000 searches, multiply only if final answer asks dollars not model units.
MCQ elimination: reject answers confusing movement/shift, legal/economic incidence, accounting/economic profit, private/social curves, or absolute/comparative advantage.

Policy trade-offs: taxes raise revenue but distort output; subsidies expand output but cost budget; quotas fix quantity but create rents; regulation can hit targets but needs information/enforcement.

Market failure	Cause	Symptom	Fix
Monopoly	market power	[illegible]	[illegible]
Externality	missing price	over[illegible]	[illegible]
Public good	free ride	unde[illegible]	[illegible]
Commons	open access	over[illegible]	[illegible]
Info	hidden type/action	adve[illegible]	[illegible]

Short answer template: define c[illegible] [illegible]iven numbers/graph; state implication; [illegible]

If asked "efficient", compare margi[illegible] [illegible]al social cost. If asked "profit-maximising", compar[illegible] [illegible]ginal cost.

Area formulas: rectangle = price/[illegible] [illegible]le = $\frac{1}{2}$ × vertical gap × quantity gap; trapezoid = av[illegible]
Common algebra errors: forgetti[illegible] [illegible]MR; using supply price as buyer price after tax; adding public-good demands horizontally; treating fixed cost as MC; using $P = ATC$ instead of $P = MC$ for competitive output.
Comparative statics sentence: "The curve shifts because ...; the movement along the other curve determines ...; therefore Q changes ..., while P is .../ambiguous."
Mini examples:

Numbers	Result
$P_D = 30 - 2Q,\ P_S = 10 + 2Q$	$Q = 5, P = 20$
+8 seller subsidy	$P_s - P_b = 8$
$P = 100 - Q,\ MC = 20$ mono	$Q_M = 40, P = 60$
same efficient	$Q_E = 80$
$MB_i = 10 - Q,\ n = 3,\ MC = 12$	$Q^* = 6$

Subsidy incidence quick solve: with $P_D = 30 - 2Q$, $P_S = 10 + 2Q$, subsidy $s = 8$: set $30 - 2Q = 10 + 2Q - 8$, so $Q = 7$, $P_b = 16$, $P_s = 24$; buyers gain 4, sellers gain 4.

For a tax/subsidy, draw two prices at the same traded quantity; do not draw separate buyer and seller quantities.

Elasticity + burden: steep demand/flat supply means buyers bear most tax; flat demand/steep supply means sellers bear most. "Steep" is visual shorthand only around equilibrium.
Perfect competition vs monopoly: comp $P = MC$ and max TS; monopoly $P > MR = MC$, restricts Q; PD may restore Q but shifts surplus.
Answering diagram questions: write the algebra even if asked to draw, because diagrams justify exact coordinates.
If options include "none": compute rather than pattern-match; ECON1001 MCQs often hide sign or incidence traps in plausible options.

Exam pacing: solve simple MCQ via concept; for calculations, write equations before numbers; for long answers, mark private vs social curves first.

High-frequency proof ideas:
Why competitive eq efficient: all units with $MB > MC$ traded and units with $MB < MC$ not traded; at Q^* marginal buyer value equals marginal seller cost.
Why monopoly DWL: monopolist ignores consumer surplus on extra units; since $MR < P$, it stops where $MR = MC$ even though $P > MC$ for additional units.
Why tax DWL: wedge prevents trades where buyer WTP exceeds seller cost by less than tax; government revenue is transfer, DWL is lost surplus.
Why public goods underprovided: each person compares private MB to price, but social MB is sum of all benefits; free riding hides true WTP.

Question phrase	Use curve	Decision rule
Willingness to pay	demand/MV	buy if $MV \ge P$
Willingness to sell	supply/MC	sell if $P \ge MC$
Social optimum	MSB/MSC	$MSB = MSC$
Private market	MPB/MPC	$MPB = MPC$
Firm output	MR/MC	$MR = MC$

Common short calculations:

Tax on sellers: if supply inverse $P_S(Q)$, new buyer price curve $P_S(Q) + t$.

Tax on buyers: new seller-received demand $P_D(Q) - t$. Same Q_t.

Subsidy to sellers: seller willing supply at buyer price shifts down by s; subsidy to buyers shifts demand up by s; same wedge.

Binding price ceiling: traded quantity is $Q_s(P_c)$; binding price floor: traded quantity is $Q_d(P_f)$ unless government buys surplus.

Surplus with discrete buyers: sort MV from high to low; efficient quantity includes units with MV at least MC; monopoly with one price compares $P \times Q$ minus cost at each possible price.
Monopoly regulation: MC pricing efficient but may make loss for natural monopoly; AC pricing covers cost but has $P > MC$ and some DWL; franchise bidding can reveal low cost but contract quality matters.
Externality instruments: Pigouvian tax fixes price of pollution; permits fix quantity of pollution; subsidies risk paying firms for baseline behavior; direct regulation needs firm cost info.
Game theory checking: dominant strategy = row-wise same best response across all columns; Nash = no unilateral profitable deviation; Pareto efficient = no other outcome makes someone better off without hurting another.
Backward induction: solve last mover's optimal action at each final node, replace subgames with payoffs, move backward to first mover.

Do not use total surplus to identify a Nash equilibrium; NE is strategic stability, not welfare maximization.

Oligopoly policy: collusion raises price and lowers quantity; antitrust targets agreements, price fixing, exclusion; some coordination may also reflect scale economies.
Elasticity signs: own-price demand elasticity negative by derivative convention; exams often use absolute value. Cross-price positive for substitutes, negative for complements; income negative for inferior goods.
Linear elasticity point: for $Q = a - bP$, at point (Q, P), $E = -bP/Q$; at demand midpoint $|E| = 1$; above midpoint elastic, below inelastic.
Welfare language: "buyers pay more" is not enough; identify CS, PS, government revenue/expenditure, and DWL.

If a policy changes who receives surplus but not quantity, efficiency may be unchanged; if quantity changes away from Q_E, DWL appears.

Final 30-sec check: signs, units, labels, which side pays/receives, private vs social, and whether the answer is asking quantity, price, surplus, profit, revenue, or welfare.

Final strategy: draw curves first, label private/social curves, then solve. For policy questions state efficiency, distribution, and implementation limits.